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# Pratik Shrivastava

Portfolio: [pratik-ruby.vercel.app](https://pratik-ruby.vercel.app)  
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## SUMMARY

Aspiring Software Engineer with a solid foundation in programming, data structures, algorithms, databases, and backend development. Experienced in building real-world applications using Python, JavaScript, FastAPI, SQL, and Git. Eager to leverage technical and analytical skills to develop scalable software solutions and deliver business impact.

## WORK EXPERIENCE

### Software Engineering Intern — Siemens Technology & Services Pvt. Ltd.

May 2024 — Aug 2024

- Developed a real-time frontend interface for an IIoT-based water monitoring platform, visualizing live consumption data from distributed sensors through interactive dashboards.
- Integrated REST APIs and implemented near real-time data updates, enabling operational monitoring and historical consumption trend analysis.
- Built GIS-based map visualizations with live meter status indicators and alerts, improving visibility into device health and flow anomalies across multiple locations.

### Data Science Intern — Dodge AI

June 2025 — July 2025

- Designed and developed an end-to-end machine learning pipeline for customer purchase behavior prediction using regression models and Random Forests.
- Built a hybrid recommendation engine integrating content-based and collaborative filtering approaches to generate personalized product recommendations.
- Conducted exploratory data analysis and engineered behavioral and transactional features to enhance prediction accuracy and uncover customer purchasing patterns.

## PERSONAL PROJECTS

### Drape

[Project Link](#)

- Built an AI-powered visual fashion search engine that finds visually similar catalog products from an uploaded photo, using **YOLOv8** for garment detection, **DINOv2** for vector embeddings, and **FAISS** for similarity search.
- Designed neighbor-consensus system that generates **NLP** garment description and retailer search links directly from retrieval results, avoiding the cost of separate captioning model.
- Deployed the **FastAPI** backend to **AWS Lambda**, to run real time inference on **20K-image** catalog within free tier limits.

### Harmonic

[Project Link](#)

- Developed a hybrid recommender blending **autoencoder** latent similarity, audio features, genre, and popularity into one ranked score, with a per-result panel explaining each signal's contribution.
- Deployed the model (**FastAPI** on Render, **React/Vite** on Vercel, **Spotify OAuth**); also cutting cold start latency by preloading model artifacts on starting using a CRON job.
- Shipped 6 surfaces on one model – recommendations, mood discovery, genre exploration, playlist building, and **3D PCA** / feature-correlation visualization of the embedding space.

### Booktunes

[Project Link](#)

- Built an AI-powered book illustration pipeline that extracts texts from PDFs via **OCR**, identifies characters using **LLM**-based **NER** with alias merging, and generates page-consistent illustrations by feeding a running visual profile in each prompt.
- Designed a layer supporting multiple **LLMs** and image generation API, including **OpenAI**, **Google Gemini**, and **Cloudflare Workers AI** via environment variables configuration directly.
- Deployed and maintained application on **Render**, **Vercel** and **Supabase PostgreSQL**.

## SKILLS

|                      |                                                                    |
|----------------------|--------------------------------------------------------------------|
| Languages            | Python, JavaScript, C++, SQL                                       |
| Software Development | OOPs, Data Structures, RESTful APIs, JWT Auth                      |
| Databases            | PostgreSQL, MySQL                                                  |
| Frameworks           | FastAPI, Node.js, Express.js                                       |
| Tools & Platforms    | Git, Github, Docker (Basics)                                       |
| ML & Data            | TensorFlow, Scikit-learn, Keras, Pandas, NumPy, OpenCV, Matplotlib |

## EDUCATION

Bachelor of Technology in Computer Science (B. Tech) — Manipal University Jaipur

July 2022 - June 2026

Relevant Coursework: Data Structures, OOPs, Software Engineering, Computer Networking, DBMS, Cloud Computing, AIML